Teacher Metacognitive Practices in Advanced Academic Courses: A Look at Bias and Pereption

A Pilot Study Conducted by

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for EDF 625: Qualitative Research
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Abstract

The Abstract is the absolute last thing you write. In the abstract, write a concise summary of the key points of your research. (Do not indent.) Your abstract clearly state that it is a pilot study and should contain at least your research topic, research questions or approach if questions haven't been identified yet, participants, methods, results, and discussion (including specific discussion with how you modify the study based on pilot data). You may also include possible implications of your research and future work you see connected with your findings. Your abstract should be a single paragraph, double-spaced. Your abstract should be between 150 and 250 words.

Example:

This pilot study used an ethnographic framework to explore the lived experiences of teachers integrating technology in a rural, Appalachian middle school. A series of participant observations of grade 3 and 5 classrooms were conducted as well as semi-structured interviews of 3 teachers and 2 administrators at the school. Thematic analysis was conducted and in vivo coding techniques were used to analyze data. Results suggest three primary themes in the data. The first theme, *We built the plane while we flew it* identifies how learning occurred through the tumultuous integration effort. The second theme, *You gotta have friends* acknowledges how the sharing the burden of the integration effort supported their learning. Finally, the third theme, *It was okay to fail* describes how feelings of administrator support made the teachers more comfortable taking risks in the classroom. Findings from the pilot informed substantive changes to the interview protocol and suggest that research questions for the regular study should focus on the area of administrator support to determine how best to encourage lasting technology integration in the classroom.

Introduction

In recent years, there has been a growing acknowledgment of the significance of metacognitive practices in the field of education, particularly in advanced academic programs like Advanced Placement (AP) and International Baccalaureate (IB). Metacognition, which refers to the awareness and self-regulation of one's learning processes, has been proven to enhance student performance by fostering deeper comprehension and improved retention of material (Owen & Vista, 2017). Despite the documented benefits, the utilization of metacognitive strategies by educators in advanced academic settings varies widely and is often influenced by factors such as school demographics and teacher perceptions. The aim of this preliminary study is to investigate the implementation of metacognitive teaching strategies among educators in advanced academic courses within different school settings, specifically, Title I schools serving underserved minority populations and schools in higher socioeconomic status (SES) areas with predominantly white student populations. The study seeks to determine whether there are significant discrepancies in the adoption and frequency of metacognitive practices between these two settings. Additionally, it aims to examine how teachers' perceptions of student capabilities may impact their utilization of these strategies.

Previous research has underscored the influence of teacher bias on student outcomes, particularly for minority and economically disadvantaged students. For example, Kao and Zimmermann (2020) discovered that teachers' perceptions of student capabilities are often influenced by race and gender, resulting in lower expectations and reduced academic support for minority students. Similarly, Leslie et al. (2015) emphasized the necessity of interventions to address teacher biases in STEM education to promote equity and enhance student performance.

This pilot study will involve observation and interviewing of educators in both Title I and higher socioeconomic status schools to assess their use of metacognitive strategies. The observations will focus on classroom practices, aiming to capture the real-time application of these strategies. The interviews, on the other hand, will delve into teachers' perceptions of their students' abilities and their reasoning behind the application or non-application of specific metacognitive techniques. By comparing these findings across different school settings, this study aims to shed light on potential disparities in metacognitive teaching practices and their implications for student learning, thereby providing valuable insights for future educational policies and practices.

Literature Review

Metacognitive Practices in Education

Metacognition is “thinking about one's thinking” and involves processes used to plan, monitor, and assess one's understanding and performance (Chick, 2023). It encompasses a critical awareness of oneself as a thinker and learner and is essential for deep learning and developing higher-order thinking skills (Taylor, 2021). Research has shown that metacognitive practices increase students' abilities to transfer or adapt their learning to new contexts and tasks by gaining awareness above the subject matter (Chen, 2020).

The Role of Metacognition in Learning

Metacognitive practices are integral to effective learning. According to Schraw and Dennison (1994), metacognitive awareness includes three components: declarative knowledge (knowledge about oneself as a learner and about what factors influence one's performance), procedural knowledge (knowledge about strategies and processes), and conditional knowledge (knowledge about when and why to use strategies). These components enable students to effectively plan, monitor, and evaluate their learning processes.

Studies have demonstrated that metacognitive strategies enhance students' problem-solving skills, decision-making abilities, and critical thinking (Pintrich, 2002). For instance, a study by Dignath and Büttner (2008) found that teaching metacognitive strategies in elementary and secondary schools significantly improved students' academic performance. Similarly, Haller, Child, and Walberg (1988) reported that metacognitive training positively influenced reading comprehension and mathematical problem-solving skills.

Metacognitive practices not only improve academic performance but also foster the development of self-regulated learning. Self-regulated learners are those who actively set goals, monitor their progress, and adjust their strategies to achieve their objectives (Zimmerman & Schunk, 2001). These learners are better equipped to handle complex and challenging tasks, demonstrating higher levels of resilience and persistence.

Teacher Bias and Its Impact on Metacognitive Practices

Implementing metacognitive practices in classrooms is significantly affected by teacher perceptions and biases. Teachers' beliefs about their students' abilities can profoundly influence their instructional methods and how much they engage students in metacognitive activities. Research indicates that in lower-income, underserved schools, teachers often face challenges impacting their ability to engage students in metacognitive work. These challenges include larger class sizes, limited resources, and the pressure to meet standardized testing benchmarks (Lehmann et al., 2022). Consequently, teachers in these settings may underestimate their students' abilities, focusing more on rote learning and test preparation rather than fostering metacognitive skills (Kao & Zimmermann, 2020; Leslie et al., 2015).

Whether implicit or explicit, teacher bias plays a crucial role in shaping classroom dynamics and instructional strategies. Studies have shown that teachers' expectations of students' abilities often align with students' socioeconomic status, race, and gender, leading to differentiated instruction that may disadvantage certain groups. For instance, Tenenbaum and Ruck (2007) found that teachers interacted more favorably with students they perceived as high achievers, often providing them with more opportunities for higher-order thinking and problem-solving activities. Conversely, students from marginalized backgrounds were less likely to be engaged in such practices, potentially limiting their academic growth and self-efficacy.

Implicit bias is unconscious attitudes or stereotypes that affect our understanding, actions, and decisions. These biases can influence teachers' expectations and interactions with students. For example, teachers may unknowingly lower their expectations for students from certain racial or socioeconomic backgrounds, which can impact their academic performance and engagement (Glock & Krolak-Schwerdt, 2013). This phenomenon underscores the importance of addressing teacher biases through professional development and reflective practices.

Disparities in Metacognitive Practices

Studies have highlighted significant disparities in using metacognitive strategies between different school settings. For instance, teachers in higher socioeconomic status (SES) schools, with predominantly white student populations are more likely to implement metacognitive practices than their counterparts in Title I schools serving underserved minority populations (Graham & Perin, 2007). This discrepancy is often attributed to the higher expectations and greater resource availability in higher SES schools, which enable teachers to focus more on metacognitive instruction (Chen, 2020).

The disparities in metacognitive practices are a matter of resource allocation but also of teacher preparedness and confidence. According to a study by Zohar and Dori (2003), teachers who received professional development in metacognitive instruction were more likely to integrate these strategies into their teaching, regardless of their school's SES. This finding suggests that targeted training and support can mitigate some of the disparities observed in different educational settings.

Furthermore, the school environment plays a critical role in shaping teachers' ability to implement metacognitive strategies. Schools with supportive leadership, collaborative professional communities, and access to instructional resources are more likely to foster environments where metacognitive practices can thrive (Fullan, 2011). Conversely, schools facing challenges such as high teacher turnover, limited professional development opportunities, and inadequate resources may struggle to sustain effective metacognitive instruction.

Impact of Teacher Training on Metacognitive Practices

Teacher training and professional development are crucial in equipping educators with the skills and knowledge necessary to implement metacognitive strategies effectively. Programs focusing on metacognitive instruction and culturally responsive pedagogy can help teachers recognize and overcome their biases, thereby improving their ability to engage all students in higher-order thinking (Owen & Vista, 2017). Studies have shown that when teachers are trained to use metacognitive strategies, they are better able to help students become aware of their strengths and weaknesses, plan their learning, and monitor their progress (Chen, 2020).

Professional development programs emphasizing reflective practice and metacognitive awareness can significantly enhance teachers' instructional methods. For instance, a study by Wilson and Bai (2010) found that teachers who engaged in professional development focused on metacognitive strategies reported increased confidence in their ability to teach these skills and observed positive changes in their students' learning behaviors. Additionally, the implementation of professional learning communities (PLCs) has been shown to support teachers in sharing best practices and collaboratively developing metacognitive instructional techniques (Vescio, Ross, & Adams, 2008).

Effective teacher training programs should also address the role of culturally responsive pedagogy in metacognitive instruction. Culturally responsive teaching practices recognize and value students' cultural backgrounds, making learning more relevant and engaging. By incorporating students' cultural references into metacognitive activities, teachers can foster a more inclusive learning environment that encourages all students to engage in higher-order thinking (Ladson-Billings, 1995).

Culturally Responsive Pedagogy and Metacognition

Culturally responsive pedagogy, which emphasizes the importance of recognizing and valuing students' cultural backgrounds in the learning process, can enhance the effectiveness of metacognitive instruction. Ladson-Billings (1995) argued that culturally responsive teaching practices not only affirm students' cultural identities but also promote higher academic achievement by making learning more relevant and engaging. By incorporating students' cultural references into metacognitive activities, teachers can foster a more inclusive learning environment that encourages all students to engage in higher-order thinking.

Research has shown that culturally responsive pedagogy can mitigate some of the negative impacts of teacher bias on student engagement. For example, a study by Gay (2002) demonstrated that teachers who adopted culturally responsive teaching practices were more likely to employ metacognitive strategies resonating with their students' lived experiences, enhancing student motivation and academic performance. Similarly, Paris and Alim (2014) highlighted the importance of culturally sustaining pedagogy, which seeks to sustain and uplift students' cultural ways of being while promoting critical and metacognitive thinking.

Implementing culturally responsive metacognitive practices involves several strategies. Teachers can create opportunities for students to reflect on their cultural identities and experiences as part of metacognitive activities. This approach validates students' backgrounds and encourages them to draw connections between their lived experiences and academic content (Hammond, 2015). Additionally, incorporating diverse perspectives and materials into the curriculum can help students develop a more comprehensive and critical understanding of the subject matter.

Metacognition and Student Outcomes

The positive impact of metacognitive practices on student outcomes is well-documented in the literature. Students who are taught to use metacognitive strategies are better equipped to understand and manage their learning processes, leading to improved academic performance across various subjects. For instance, a meta-analysis by Hattie (2009) found that metacognitive strategies had a significant effect size of 0.69 on student achievement, indicating their substantial impact on learning outcomes.

Moreover, the benefits of metacognitive practices extend beyond academic performance. Students who develop strong metacognitive skills are likelier to exhibit greater self-efficacy, resilience, and motivation (Zimmerman & Schunk, 2001). These skills are critical for lifelong learning and success in various domains, including higher education and the workforce. For example, a study by Dignath and Büttner (2008) found that elementary and secondary students who received metacognitive training showed increased motivation and persistence in challenging tasks, suggesting that metacognitive practices foster a growth mindset and positive attitudes toward learning.

Metacognitive practices also play a crucial role in fostering critical thinking and problem-solving skills. Students who engage in metacognitive activities are more likely to analyze and evaluate information critically, leading to better decision-making and problem-solving abilities (Kuhn & Dean, 2004). This is particularly important in today's rapidly changing world, where the ability to think critically and adapt to new situations is essential for success.

Challenges and Barriers to Implementing Metacognitive Practices

Despite the well-documented benefits of metacognitive practices, several challenges and barriers hinder their widespread implementation in classrooms. One significant barrier is the lack of teacher training and support. Many teachers report feeling unprepared to teach metacognitive strategies and express a need for more professional development opportunities (Wilson & Bai, 2010). The pressure to meet standardized testing requirements can also limit teachers' ability to incorporate metacognitive instruction into their curricula, particularly in under-resourced schools.

Another challenge is the potential for teacher bias to influence implementing metacognitive practices. As previously discussed, teachers' perceptions of their students' abilities can shape their instructional methods, potentially leading to lower expectations and reduced opportunities for higher-order thinking for certain student groups (Kao & Zimmermann, 2020). Addressing these biases through professional development and culturally responsive pedagogy is essential for ensuring that all students have access to metacognitive instruction.

Additionally, systemic issues such as inadequate funding, high student-teacher ratios, and lack of access to instructional resources can impede the implementation of metacognitive practices. Schools in lower-income areas often face these challenges, making it difficult for teachers to provide the individualized support necessary for effective metacognitive instruction (Darling-Hammond, 2010).

Strategies for Overcoming Barriers

Several strategies can be employed to support the implementation of metacognitive practices in classrooms to overcome these challenges. First, comprehensive and ongoing professional development programs should be provided to equip teachers with the necessary skills and knowledge to teach metacognitive strategies effectively. These programs should include training on reflective practices, culturally responsive pedagogy, and methods for fostering self-regulated learning (Fullan, 2011).

Second, schools should establish professional learning communities (PLCs) where teachers can collaborate, share best practices, and develop metacognitive instructional techniques together. PLCs can provide a supportive environment for teachers to learn from one another and continuously improve their instructional methods (Vescio, Ross, & Adams, 2008).

Third, policymakers and educational leaders should advocate for equitable funding and resource allocation to ensure that all schools, particularly those in underserved areas, have the necessary resources to support metacognitive instruction. This includes providing access to instructional materials, technology, and support staff to assist with implementing metacognitive practices (Darling-Hammond, 2010).

Finally, fostering a school culture that values and prioritizes metacognitive practices is essential. School leaders should encourage teachers to integrate metacognitive activities into their daily instruction and provide opportunities for students to reflect on their learning processes regularly. Celebrating successes and showcasing examples of effective metacognitive instruction can help build a culture of reflective practice within the school community (Hattie, 2009).

Statement of Problem

While the literature review shows that research in the area of metacognitive practices in education has extensively addressed the importance of metacognition in enhancing student performance and self-regulated learning (Flavell, 1979; Schraw & Moshman, 1995; Zimmerman & Schunk, 2001), there is little research that focuses specifically on the implementation of these practices in Advanced Placement (AP) programs within underserved school populations. Existing studies have highlighted the disparities in the application of metacognitive strategies across different socioeconomic contexts, with a significant gap in resources and teacher preparedness affecting underserved schools (Graham & Perin, 2007; Chen, 2020; Lehmann et al., 2022). Moreover, the impact of teacher biases on the engagement of students in metacognitive activities has been identified as a critical barrier to the equitable application of these strategies (Tenenbaum & Ruck, 2007; Glock & Krolak-Schwerdt, 2013).

This pilot study aims to explore and test methods meant to help determine how best to study the gap in the literature related to the implementation of metacognitive practices in AP programs, particularly in underserved schools. By focusing on teacher training and culturally responsive pedagogy, this study will seek to understand how to effectively engage students from marginalized populations in metacognitive activities and assess the impact on their academic performance. The findings from this pilot study will provide insights into developing comprehensive strategies for integrating metacognitive practices in AP courses, addressing both the instructional and perceptual barriers that currently exist.

Methods

The study uses a mixed-methods approach, combining qualitative observations and interviews with quantitative data analysis. Specifically, it utilizes a phenomenological study design to explore the lived experiences and perceptions of teachers regarding their use of metacognitive teaching strategies. Phenomenological research focuses on understanding how individuals perceive and make sense of their experiences, making it suitable for this pilot study, which aims to delve into teachers' beliefs and practices (Moustakas, 1994).

Phenomenology is particularly appropriate for this study because it seeks to understand the essence of teachers' experiences with metacognitive strategies in different educational contexts. By capturing detailed accounts of their practices and perceptions, this approach allows for a deep exploration of the factors influencing their instructional methods.

Participants

The study will involve teachers from advanced academic courses at two types of schools: Title I schools with a predominantly minority and low SES student population, and higher SES schools with a predominantly white student population. A total of 20 teachers (10 from each type of school) will be selected based on their experience and willingness to participate. This selection aims to ensure a diverse representation of teaching practices and school environments (Graham & Perin, 2007).

Procedures

Observations: Classroom observations will be conducted to assess the frequency and nature of metacognitive teaching strategies used by teachers. An observation protocol will be developed to systematically record instances of metacognitive practices, such as prompting students to reflect on their learning, encouraging self-assessment, and using metacognitive prompts. These observations will help identify how often and in what ways metacognitive strategies are integrated into daily instruction (Angelo & Cross, 2012).

Interviews: Semi-structured interviews will be conducted with each teacher to explore their perceptions of student abilities and their use of metacognitive strategies. The interview questions will focus on teachers' beliefs about the effectiveness of these strategies, their experiences with different student populations, and any barriers they face in implementing metacognitive practices. This method allows for in-depth exploration of teachers'

Reflexivity/Ethics

Reflexivity is crucial in qualitative research to acknowledge the researcher's role and potential biases in the study. As a researcher, I will maintain a reflexive journal to document my thoughts, feelings, and reactions throughout the data collection and analysis process. This practice helps to identify and mitigate any biases that may influence the interpretation of the data (Saldana, 2011). Ethical considerations are paramount in this study. All participants will provide informed consent before participating in the study. Confidentiality will be maintained by anonymizing the data, and participants will have the right to withdraw from the study at any time. Ethical approval will be obtained from the relevant institutional review board (IRB) before data collection begins (Saldana, 2011).

Expected Outcomes

This study aims to uncover potential disparities in the use of metacognitive strategies between teachers in different school settings. It is expected that teachers in higher SES schools may employ these strategies more frequently, given the typically higher expectations and resource availability. Conversely, teachers in Title I schools may face challenges that hinder their use of metacognitive practices. The findings will provide insights into how to better support teachers in implementing effective metacognitive strategies across diverse educational contexts (Graham & Perin, 2007).

Results

Coding Methods

The data for this pilot study was analyzed using thematic coding, a method recommended by Saldana (2011) for its effectiveness in qualitative research. Thematic coding involves identifying patterns or themes within qualitative data, which can then be used to describe the data set comprehensively. This approach was chosen because it allows for a detailed understanding of teachers' use of metacognitive strategies and their perceptions of student abilities across different school settings. The coding process involved reading through the observation notes and interview transcripts multiple times to identify recurring themes. These themes were then categorized and analyzed to provide a narrative explanation of the findings.

Engagement and Interactive Strategies

Classroom observations highlighted stark contrasts in student engagement between the instructional practices of Mr. B and Mr. C. Mr. C’s use of traditional, passive instructional methods such as direct lectures and presentations led to low levels of student interaction. For instance, during an AP Human Geography lesson, Mr. C's attempt to engage students through discussion resulted in minimal participation, with many students distracted and disinterested. Mr. C noted, "I use a lot of group work and, uh, projects. We also watch videos and use maps. I think these methods work, but... sometimes the students don't seem that interested" (Interview Transcript #1). This lack of engagement was further highlighted during reflective journaling activities, where students' entries were often brief and superficial, indicating a shallow level of cognitive engagement. Mr. C remarked, "Well, during the urbanization unit, I had them keep journals about their thoughts and questions. We discussed these in class. It was okay, but... I think it could have been better" (Interview Transcript #1).

The more dynamic teaching approach by Mr. B fostered a highly engaging classroom environment. His use of interactive activities such as debates, group projects, and simulations encouraged students to participate and apply critical thinking skills actively. A notable example was a simulation of the Congress of Vienna, where students, acting as diplomats, negotiated treaties and alliances. Mr. B explained, "We did a simulation of the Congress of Vienna where students took on roles of different historical figures. They had to negotiate and agree, which helped them understand the complexities of diplomacy and historical context" (Interview Transcript #2). Mr. B's reflective journaling and Socratic seminars further supported critical thinking and self-assessment, as students posed thoughtful questions and engaged in meaningful discussions. "I incorporate reflective journaling and Socratic seminars where students discuss and question each other's ideas. These activities help students internalize their learning and develop critical thinking skills" (Interview Transcript #2). These observations suggest that interactive and student-centered teaching methods significantly enhance student engagement, in contrast to the more passive techniques observed in Mr. C's class.

Challenges in Implementing Metacognitive Practices

The implementation of metacognitive strategies also varied significantly between the two schools. Mr. C, who was instructing students in the AP Summer Bootcamp at Polytechnic High School, faced notable challenges in this area. His attempts to integrate reflective journaling into lessons, such as during a unit on urbanization, were met with limited success. Students often produced shallow reflections and required frequent reminders to stay focused, reflecting a lack of familiarity and comfort with metacognitive practices. Mr. C expressed, "I try to have them write in journals and we do some class discussions. But, um, getting them to really think deeply... that's been challenging. I'm not sure I'm doing it right" (Interview Transcript #1). Mr. C attributed some of these challenges to the socio-economic difficulties faced by his students, which he believed affected their ability to engage deeply with the material. "Many of my students have a lot going on outside of school, and it's hard for them to concentrate fully" (Interview Transcript #1).

Using more effective metacognitive strategies, Mr. B, who was teaching at the AP Bootcamp at Dunbar High School, instructed in a generally more cognitively rigorous manner. His confidence in implementing reflective practices, likely influenced by his experience at a higher SES school, led to richer cognitive engagement among students. During Socratic seminars, students engaged deeply with the material, asking insightful questions and challenging each other's viewpoints. Mr. B detailed, "During our unit on revolutions, I had students keep a journal where they reflected on each revolution's causes, events, and outcomes. We then used these reflections in a class debate, which helped them articulate and deepen their understanding" (Interview Transcript #2). This approach not only fostered critical thinking but also encouraged students to take ownership of their learning. "I incorporate reflective journaling and Socratic seminars where students discuss and question each other's ideas. These activities help students internalize their learning and develop their critical thinking skills" (Interview Transcript #2). The disparity in the effectiveness of metacognitive practices highlights the importance of teacher confidence and experience in fostering deeper cognitive engagement among students.

Teacher Perceptions and Biases

Teacher perceptions and biases also played a crucial role in shaping instructional practices. Mr. C expressed uncertainty about his ability to engage students effectively, which may have contributed to the lower levels of student engagement and the superficial implementation of metacognitive practices at Polytechnic High School. Mr. C admitted, "I'm not sure if I'm reaching them the way I should be" (Interview Transcript #1). His perception of the socio-economic challenges faced by his students seemed to influence his teaching approach, potentially limiting his effectiveness. Mr. C noted, "It's tough when you know they're dealing with so much outside of school. It’s hard to get them to focus on metacognition" (Interview Transcript #1).

Mr. B demonstrated a strong belief in the potential of all students, regardless of their background. His culturally responsive teaching methods, which included incorporating diverse perspectives and fostering a collaborative learning environment, reflected his commitment to supporting marginalized students. Mr. B stated, "I include diverse perspectives and examples in my lessons. I also make an effort to learn about my students’ backgrounds and incorporate their experiences into our discussions to make the material more relevant to them" (Interview Transcript #2). This positive perception likely contributed to the higher levels of student engagement and the effective implementation of metacognitive strategies observed in his classroom. "When students see their own experiences reflected in the material, they are more engaged and motivated to learn" (Interview Transcript #2). These findings underscore the significant impact of teacher perceptions and biases on student engagement and the effectiveness of metacognitive practices. Addressing these biases through targeted professional development can help teachers better support diverse student populations.

Discussion

This is where you take your results and discuss what they might mean and/or their implications. Discussions can focus on any or all of the following:

1. why you might have gotten the results you did,
2. what you found that was unexpected or weird,
3. what do you (or others) need to do further research on,

4. how this info might be use to modify practice.

In addition to this, for this pilot you will be asked to include the following subsection in which you discuss how conducting the pilot gave you ideas for how you would modify your study

Modifications to Study Based on Pilot Results

Use this section to describe how the results of the pilot study might inform the design of a real, larger scale study. What you would do differently? How would you modify your interview schedule or observation techniques? Did your research help you better determine what your research questions should be? Should you modify your procedure? How should you sample participants if you do this on a larger scale? Who else should you obtain interviews/observations from?

Length: 3-5 pages, , Times New Roman, 12 point, double spaced.

Class Reflection

This section is the general reflection on the class. Include what you learned, what you struggled with, and how you would like to see the class improved.

Length: 1 page, Times New Roman, 12 point, double spaced.

References

Put your APA citations here in alphabetical order.

Appendix A: Permission to Access Your Population

Attach a copy of the email or written permission you received to access your population.

Appendix B: Pilot Interview Schedule

Include your original interview schedule.

Appendix C: Proposed Revision to Interview Schedule

Include how you would revise the interview schedule based on the results of your interview.

Appendix D: Informed Consent

Attach copies of your informed consent letter showing your participant read and signed off on the interview.

Appendix E: Codebook

Please provide a copy of your codebook with your big themes, subcodes and definitions for each.